## Table 1. Data sources and their roles

| Dataset / endpoint | Cases | Controls | Role |
|---|---|---|---|
| eQTLGen blood cis-eQTL | — | — | Exposure; up to 31,684 participants (median 28,092 per variant); predominantly European |
| BBJ GD (GCST90018627) | 2,809 | 172,656 | Discovery; East Asian |
| FinnGen GD (E4_GRAVES_STRICT) | 3,962 | 496,386 | Post hoc same-disease comparison; Finnish |
| UKB hyperthyroidism (GCST90038636) | 3,731 | 480,867 | Broader-phenotype comparison; self-reported hyperthyroidism or thyrotoxicosis; ancestry not established as exclusively European |
| FinnGen GO (GRAVES_OPHT) | 858 | 499,490 | Original ophthalmopathy (TED-enriched) comparison; Finnish |
| FinnGen GO strict (E4_GRAVES_OPHT_STRICT) | 753 | 499,595 | Post hoc ophthalmopathy definition sensitivity; Finnish |
| GTEx v8 tissue eQTL | — | — | Post hoc tissue extension (thyroid, cultured fibroblasts, subcutaneous and visceral adipose, whole blood; n = 469–670; Table S4) |
| 1000 Genomes Phase 3 European reference (n = 503) | — | — | LD clumping of instruments and reference allele frequencies |

The three FinnGen R12 endpoints share participants and are not independent replications. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. UKB hyperthyroidism is broader than GD. ‘Strict’ is the FinnGen endpoint label, not proof of perfect specificity. The descriptive orbital RNA-sequencing series (four TED patients, one control) is separate from these datasets. eQTL, expression quantitative trait locus; LD, linkage disequilibrium; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table 2. MR and colocalization results for *TSHR*, *IGF1R* and *CTLA4* across five disease datasets

| Gene | Dataset | IVs | OR (95% CI) | *P* value | Q-test *P* | PP.H2 | PP.H3 | PP.H4 | PP.H4, p12 = $10^{-6}$ |
|---|---|---|---|---|---|---|---|---|---|
| *TSHR* | BBJ GD | 1 | 0.12 (0.07–0.21) | $1.09\times10^{-14}$*† | NA | <0.001 | 0.049 | **0.951** | 0.661 |
| *TSHR* | FinnGen GD | 1 | 0.08 (0.05–0.12) | $3.19\times10^{-32}$*† | NA | <0.001 | 0.736 | 0.264 | 0.035 |
| *TSHR* | UKB hyperthyroidism | 1 | 0.09 (0.06–0.14) | $8.77\times10^{-28}$*† | NA | <0.001 | 0.774 | 0.226 | 0.028 |
| *TSHR* | FinnGen GO | 1 | 0.10 (0.04–0.24) | $2.82\times10^{-7}$*† | NA | <0.001 | 0.014 | **0.986** | **0.875** |
| *TSHR* | FinnGen GO strict | 1 | 0.15 (0.06–0.39) | $9.02\times10^{-5}$* | NA | 0.027 | 0.082 | **0.891** | 0.451 |
| *IGF1R* | BBJ GD | 4 | 1.56 (1.07–2.28) | 0.0212* | 0.902 | 0.690 | 0.236 | 0.073 | 0.008 |
| *IGF1R* | FinnGen GD | 3 | 1.07 (0.86–1.33) | 0.539 | 0.882 | 0.731 | 0.259 | 0.010 | 0.001 |
| *IGF1R* | UKB hyperthyroidism | 4 | 1.35 (1.07–1.70) | 0.0117* | 0.250 | 0.400 | 0.196 | 0.404 | 0.063 |
| *IGF1R* | FinnGen GO | 3 | 1.41 (0.85–2.33) | 0.182 | 0.309 | 0.623 | 0.346 | 0.032 | 0.003 |
| *IGF1R* | FinnGen GO strict | 3 | 1.52 (0.92–2.49) | 0.0996 | 0.409 | 0.571 | 0.391 | 0.038 | 0.004 |
| *CTLA4* | BBJ GD | 1 | 0.18 (0.11–0.27) | $5.45\times10^{-15}$*† | NA | <0.001 | 0.799 | 0.201 | 0.025 |
| *CTLA4* | FinnGen GD | 2 | 0.17 (0.08–0.36) | $2.15\times10^{-6}$*† | 0.0275 | <0.001 | 0.746 | 0.254 | 0.033 |
| *CTLA4* | UKB hyperthyroidism | 2 | 0.21 (0.08–0.57) | 0.00220* | 0.00131 | <0.001 | 0.047 | **0.953** | 0.672 |
| *CTLA4* | FinnGen GO | 2 | 0.17 (0.04–0.68) | 0.0118* | 0.0486 | <0.001 | 0.022 | **0.978** | **0.816** |
| *CTLA4* | FinnGen GO strict | 2 | 0.17 (0.05–0.64) | 0.00888* | 0.0755 | <0.001 | 0.021 | **0.979** | **0.820** |

MR estimates are ORs per approximately one SD higher genetically predicted blood expression: Wald ratio for one IV, multiplicative random-effects IVW for several. *TSHR* used one instrument, rs179252, in every dataset; its G allele raises blood expression by 0.105 SD per allele (F = 176), and its per-allele disease ORs were 0.77–0.82 (Table S3). The per-SD ORs therefore rescale a moderate per-allele association, and each single-IV *P* value equals the variant's disease-association z-test *P* value. *CTLA4* multi-IV estimates are concentrated in rs13030124 and heterogeneous (Q-test *P*); in BBJ and UKB, two of four single-instrument omissions removed nominal *IGF1R* significance (Figure S2). * Nominal $P < 0.05$; † $P < 0.05/2{,}544$, the screen threshold, applied descriptively outside BBJ. PP.H2–H4 are colocalization posterior probabilities (H2, expression signal only; H3, distinct variants; H4, one shared variant) with $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$ unless stated; bold, ≥0.80. They are not *P* values. BBJ colocalization pairs European eQTL with East Asian disease statistics. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; MR, Mendelian randomization; NA, not applicable (one IV); OR, odds ratio; PP, posterior probability; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table 3. Thirteen BBJ discoveries and their FinnGen GD comparison

| BBJ discovery gene | Chr | IVs, BBJ/FinnGen GD | BBJ OR (95% CI) | BBJ *P* | FinnGen GD OR (95% CI) | FinnGen GD *P* | Focused support | FinnGen GD PP.H4 |
|---|---|---|---|---|---|---|---|---|
| *HLA-A* | 6 (MHC) | 4/4 | 1.98 (1.73–2.27) | $2.60\times10^{-23}$ | 1.08 (0.77–1.53) | 0.648 | No | <0.001 |
| *HLA-DQA2* | 6 (MHC) | 1/1 | 2.39 (1.95–2.93) | $8.30\times10^{-17}$ | 1.90 (1.64–2.20) | $1.41\times10^{-17}$ | Yes | <0.001 |
| *CTLA4* | 2 | 1/2 | 0.18 (0.11–0.27) | $5.45\times10^{-15}$ | 0.17 (0.08–0.36) | $2.15\times10^{-6}$ | Yes | 0.254 |
| *TSHR* | 14 | 1/1 | 0.12 (0.07–0.21) | $1.09\times10^{-14}$ | 0.08 (0.05–0.12) | $3.19\times10^{-32}$ | Yes | 0.264 |
| *C4A* | 6 (MHC) | 1/3 | 0.45 (0.35–0.57) | $2.23\times10^{-10}$ | 0.41 (0.26–0.63) | $4.97\times10^{-5}$ | Yes | <0.001 |
| *HSD3B7* | 16 (16p11.2 cluster) | 1/2 | 0.29 (0.18–0.47) | $2.04\times10^{-7}$ | 0.90 (0.72–1.13) | 0.372 | No | 0.011 |
| *TUBB* | 6 (MHC) | 2/3 | 1.49 (1.28–1.74) | $3.14\times10^{-7}$ | 3.28 (2.09–5.17) | $2.68\times10^{-7}$ | Yes | <0.001 |
| *VKORC1* | 16 (16p11.2 cluster) | 1/0 | 0.13 (0.06–0.29) | $6.35\times10^{-7}$ | NA | NA | NA | 0.011 |
| *TNFSF14* | 19 | 1/1 | 0.63 (0.52–0.76) | $1.48\times10^{-6}$ | 0.77 (0.65–0.92) | 0.00361 | Yes | 0.303 |
| *PRSS36* | 16 (16p11.2 cluster) | 1/1 | 6.00 (2.87–12.57) | $2.02\times10^{-6}$ | 1.40 (1.00–1.96) | 0.0490 | No | 0.063 |
| *MAPKAPK5* | 12 | 4/4 | 0.977 (0.967–0.987) | $5.28\times10^{-6}$ | 0.990 (0.929–1.055) | 0.751 | No | 0.041 |
| *PSMB8* | 6 (MHC) | 1/1 | 1.25 (1.13–1.38) | $6.86\times10^{-6}$ | 0.96 (0.85–1.08) | 0.512 | No | <0.001 |
| *IFNGR1* | 6 | 1/1 | 2.10 (1.51–2.91) | $9.41\times10^{-6}$ | 1.62 (1.18–2.24) | 0.00330 | Yes | 0.313 |

Genes passed the BBJ screen threshold ($P < 0.05/2{,}544$) and are listed by BBJ *P* value. Focused support in FinnGen GD requires the same direction and $P < 0.05/13$; 7 of 13 qualify. IVs are counted separately for each dataset because instrument availability differs. The five MHC genes and the three chromosome 16p11.2 genes are not independent signals; the latter are compatible with one regional association. *MAPKAPK5* is shown to three decimals to preserve its small effect. NA, not estimable (no retained instrument), not a negative result; *VKORC1* colocalization does not require an MR instrument. PP.H4, shared-variant posterior probability under $p12 = 10^{-5}$; bold, ≥0.80. Chr, chromosome of the strongest instrument; MHC, major histocompatibility complex; CI, confidence interval; IV, instrumental variable; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Figure 1

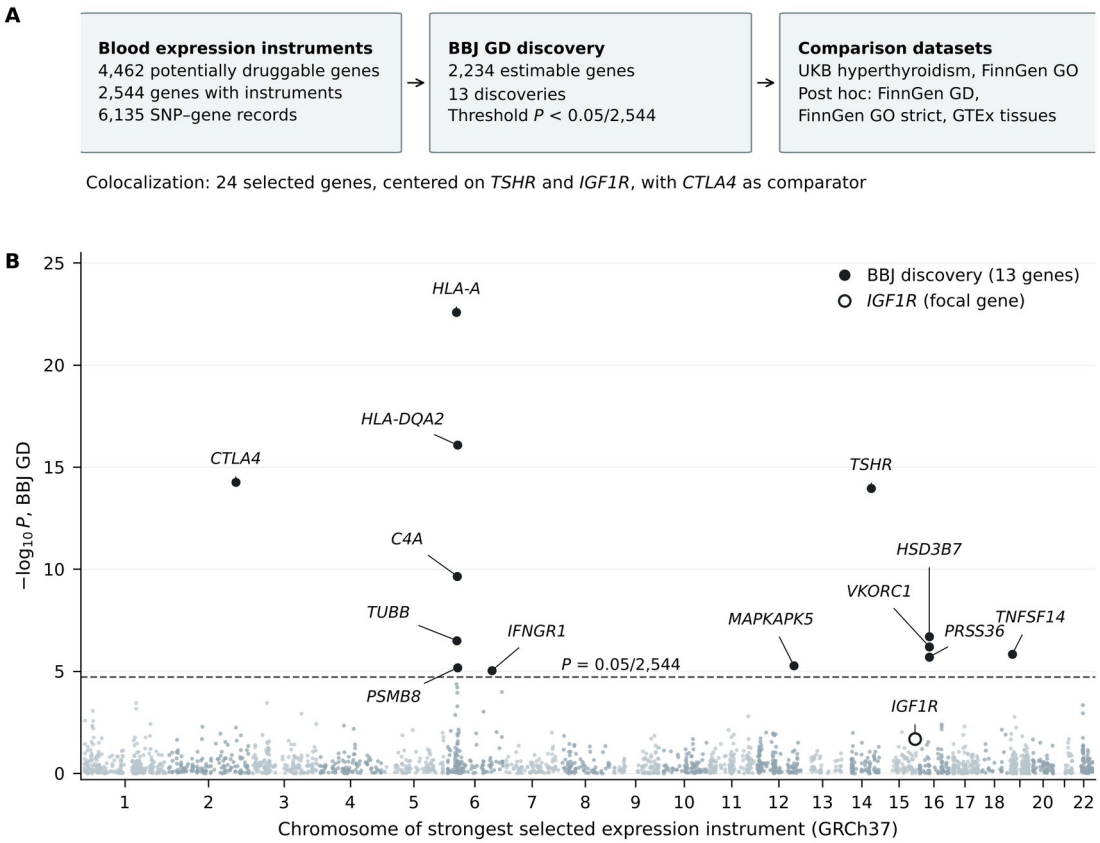



**Figure 1. Study design and gene-level discovery in BBJ.** (A) Analysis flow. Blood expression instruments from eQTLGen were tested against BBJ GD for discovery; UKB hyperthyroidism and FinnGen GO were the original comparison datasets, and FinnGen GD, FinnGen GO strict and the GTEx tissue extension were added post hoc. The three FinnGen endpoints overlap and are not independent replications. (B) Each grey point is one estimable gene (n = 2,234), placed at the GRCh37 position of its strongest selected expression instrument and plotted by its BBJ MR *P* value; points are genes, not single variants. Filled black points are the 13 genes that passed $P < 0.05/2{,}544$ (dashed line, $-\log_{10}P \approx 4.71$); the open circle is the focal gene *IGF1R*, which did not. The threshold applies to the BBJ discovery screen, not to later exploratory analyses. GTEx, Genotype-Tissue Expression; MR, Mendelian randomization; SNP, single-nucleotide polymorphism; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Figure 2

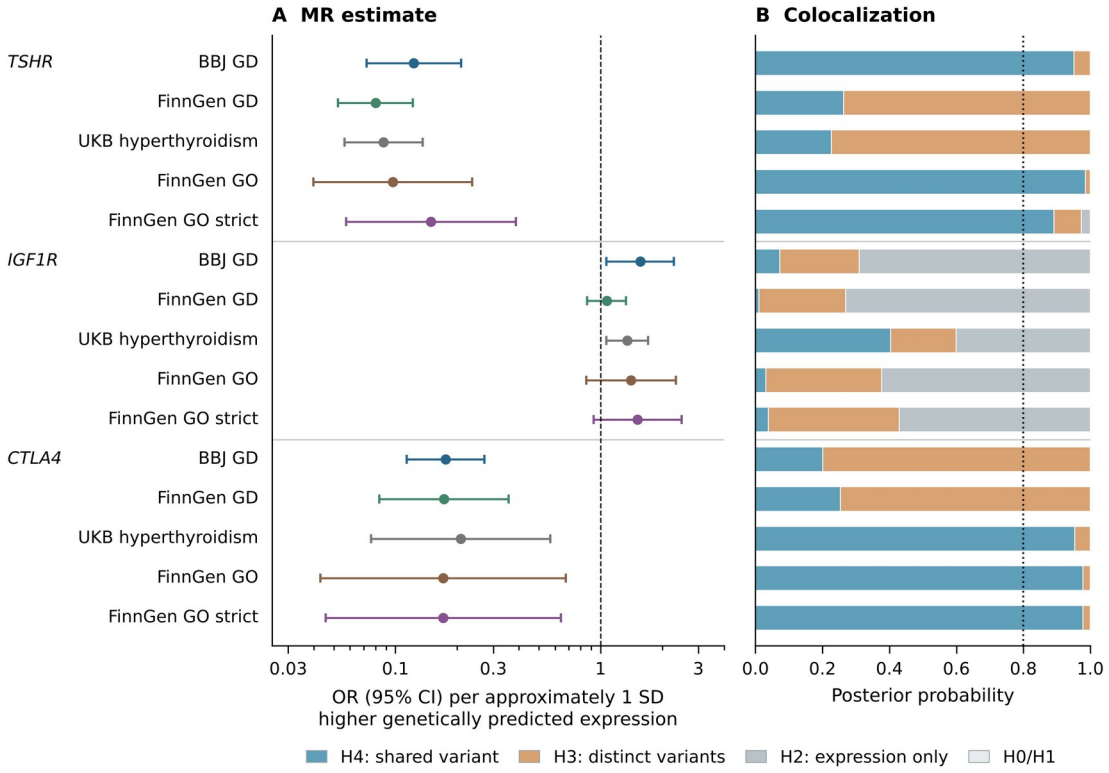



**Figure 2. *TSHR*, *IGF1R* and *CTLA4* across five disease datasets.** (A) MR odds ratios with 95% CIs per approximately 1 SD higher genetically predicted blood expression: Wald ratio for one instrument, multiplicative random-effects IVW for several. The axis is logarithmic and the dashed line is OR = 1. *TSHR* used one instrument, rs179252, in every dataset. Colours identify datasets. (B) Colocalization posterior probabilities for the same comparisons ($p1 = p2 = 10^{-4}$, $p12 = 10^{-5}$), stacked from zero as H4 (one shared variant), H3 (distinct variants), H2 (expression signal only) and H0/H1 (neither trait, or disease only). The dotted line marks PP.H4 = 0.80, the descriptive reporting threshold. Estimates, *P* values and posterior probabilities are given in Table 2; *P* values and posterior probabilities answer different questions. FinnGen GO controls include people with GD but no recorded eye disease. CI, confidence interval; IVW, inverse-variance weighted; MR, Mendelian randomization; OR, odds ratio; PP, posterior probability; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Figure 3

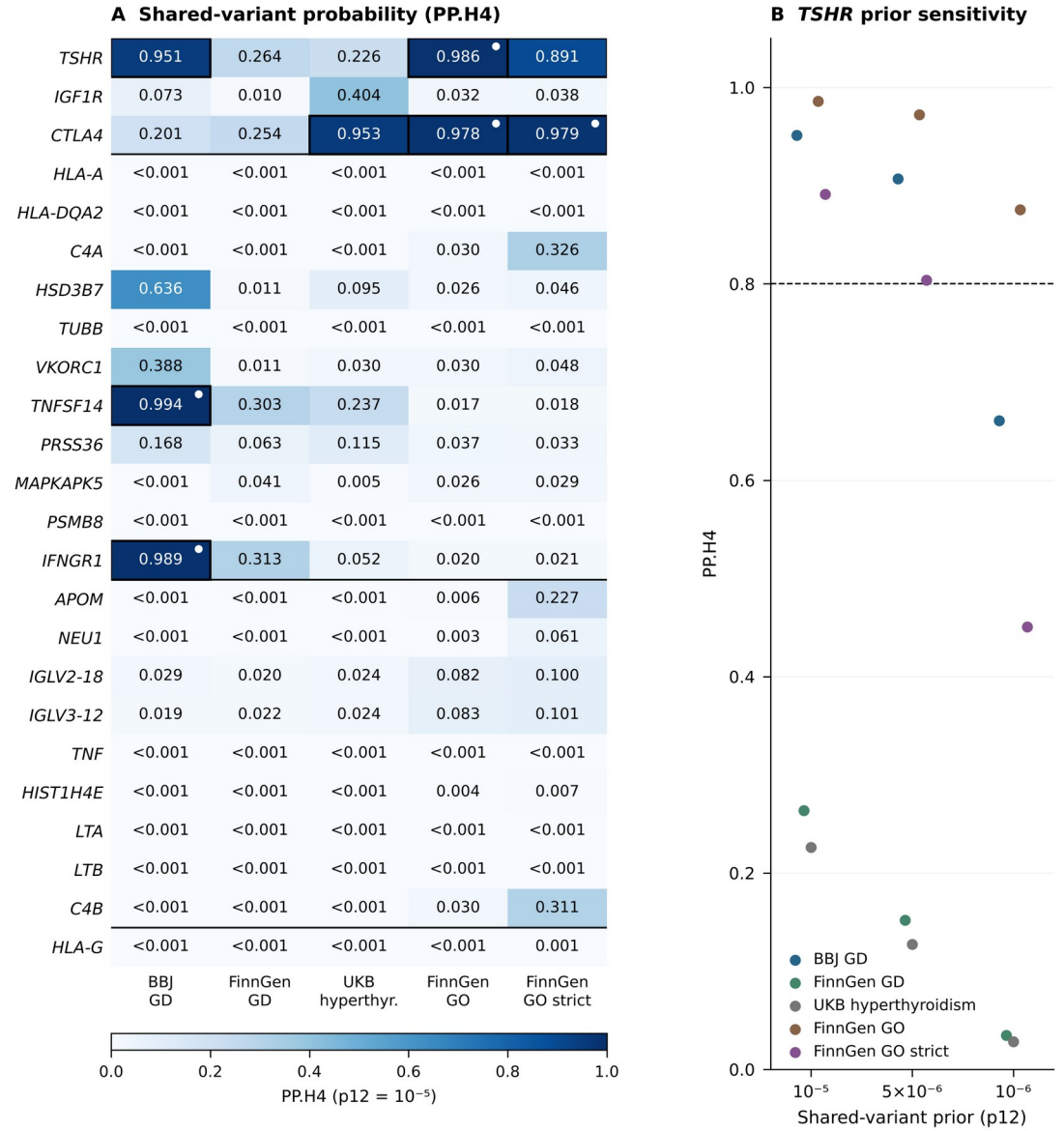



**Figure 3. Colocalization across 24 selected genes and *TSHR* prior sensitivity.** (A) PP.H4 for each gene and dataset (p1 = p2 = $10^{-4}$, p12 = $10^{-5}$); shading follows the colour bar. Boxed cells have PP.H4 ≥ 0.80, and a dot marks cells that remain ≥ 0.80 at p12 = $10^{-6}$. Horizontal lines separate *TSHR*, *IGF1R* and *CTLA4*; the other 11 BBJ discoveries; the nine additional FinnGen GD discoveries; and *HLA-G*, added because its MR result changed with the allele-frequency source. Only these 24 genes underwent colocalization. Values below 0.001 are shown as <0.001; full-precision values are in the Supplementary Dataset. Low BBJ PP.H4 for the two *IGLV* genes partly reflects absence of the lead expression variant from the BBJ data. (B) *TSHR* PP.H4 in each dataset at three shared-variant priors (p12); points are offset slightly to avoid overlap, and the dashed line marks 0.80. PP.H4 is a posterior probability under the colocalization model, not a *P* value. PP.H4, posterior probability of one shared causal variant; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S1. Additional FinnGen GD screen discoveries

| Gene | Chr | FinnGen GD IVs | FinnGen GD OR (95% CI) | FinnGen GD *P* | BBJ PP.H4 | FinnGen GD PP.H4 | FinnGen GO strict PP.H4 |
|---|---|---|---|---|---|---|---|
| *APOM* | 6 (MHC) | 1 | 0.05 (0.04–0.06) | $1.43\times10^{-108}$ | <0.001 | <0.001 | 0.227 |
| *NEU1* | 6 (MHC) | 1 | 0.04 (0.03–0.06) | $1.43\times10^{-108}$ | <0.001 | <0.001 | 0.061 |
| *IGLV2-18* | 22 | 1 | 0.56 (0.48–0.66) | $7.52\times10^{-12}$ | 0.029 | 0.020 | 0.100 |
| *IGLV3-12* | 22 | 1 | 1.20 (1.14–1.26) | $7.52\times10^{-12}$ | 0.019 | 0.022 | 0.101 |
| *TNF* | 6 (MHC) | 1 | 0.60 (0.51–0.71) | $4.65\times10^{-10}$ | <0.001 | <0.001 | <0.001 |
| *HIST1H4E* | 6 (MHC) | 1 | 2.54 (1.88–3.43) | $1.51\times10^{-9}$ | <0.001 | <0.001 | 0.007 |
| *LTA* | 6 (MHC) | 1 | 0.46 (0.35–0.61) | $3.75\times10^{-8}$ | <0.001 | <0.001 | <0.001 |
| *LTB* | 6 (MHC) | 1 | 0.18 (0.10–0.36) | $4.89\times10^{-7}$ | <0.001 | <0.001 | <0.001 |
| *C4B* | 6 (MHC) | 2 | 1.63 (1.31–2.03) | $1.10\times10^{-5}$ | <0.001 | <0.001 | 0.311 |

Genes outside the BBJ discovery set that passed $P < 0.05/2{,}544$ in the post hoc full FinnGen GD screen (1000 Genomes European reference frequencies). These are not independent of the screen and seven of nine are in the MHC. *APOM*/*NEU1* share rs3101018 and *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. None reached PP.H4 ≥0.80 ($p12 = 10^{-5}$) in any dataset. Chr, chromosome; IV, instrumental variable; MHC, major histocompatibility complex; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S2. Frequency-sensitive exploratory candidates

| Gene | Dataset | Frequency source | IVs | OR (95% CI) | *P* value | Q-test *P* |
|---|---|---|---|---|---|---|
| *C4B* | FinnGen GD | 1000G EUR | 2 | 1.63 (1.31–2.03) | $1.10\times10^{-5}$ | $7.79\times10^{-7}$ |
| *C4B* | FinnGen GD | eQTLGen cohort | 2 | 1.75 (1.31–2.33) | $1.48\times10^{-4}$ | $1.36\times10^{-8}$ |
| *C4B* | FinnGen GO strict | 1000G EUR | 2 | 1.68 (1.44–1.95) | $2.74\times10^{-11}$ | 0.122 |
| *C4B* | FinnGen GO strict | eQTLGen cohort | 2 | 1.80 (1.46–2.23) | $6.17\times10^{-8}$ | 0.0587 |
| *HLA-G* | FinnGen GD | 1000G EUR | 16 | 0.86 (0.72–1.03) | 0.107 | $9.98\times10^{-24}$ |
| *HLA-G* | FinnGen GD | eQTLGen cohort | 16 | 0.60 (0.47–0.75) | $1.72\times10^{-5}$ | $2.26\times10^{-10}$ |
| *HLA-G* | FinnGen GO strict | 1000G EUR | 16 | 0.91 (0.76–1.08) | 0.267 | 0.0231 |
| *HLA-G* | FinnGen GO strict | eQTLGen cohort | 16 | 0.70 (0.53–0.92) | 0.0122 | 0.131 |

Exposure effects were reconstructed with 1000 Genomes European reference allele frequencies (primary) or eQTLGen cohort frequencies; variant sets and outcome effects were held fixed, so changes reflect rescaled expression effects and IVW weights. *HLA-G* was added because its eQTLGen-frequency *P* crossed the screen threshold ($P < 0.05/2{,}544$). Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are in the workbook. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S3. *TSHR* regional colocalization and prior sensitivity

| Dataset | Matched variants | Disease lead (*P*) | rs179252 *P* | rs179252-G OR (95% CI) | H4 top variant | PP.H4, $10^{-5}$ / $10^{-6}$ | PP.H4 range across windows |
|---|---|---|---|---|---|---|---|
| BBJ GD | 4,336 | rs28414437 ($1.83\times10^{-15}$) | $1.09\times10^{-14}$ | 0.80 (0.76–0.85) | rs179252 | **0.951** / 0.661 | 0.951–0.951 |
| FinnGen GD | 6,579 | rs1023586 ($1.58\times10^{-36}$) | $3.19\times10^{-32}$ | 0.77 (0.73–0.80) | rs1023586 | 0.264 / 0.035 | 0.264–0.264 |
| UKB hyperthyroidism | 5,667 | rs4903961 ($1.17\times10^{-43}$) | $8.77\times10^{-28}$ | 0.77 (0.74–0.81) | rs1023586 | 0.226 / 0.028 | 0.226–0.226 |
| FinnGen GO | 6,579 | rs3783947 ($1.23\times10^{-7}$) | $2.82\times10^{-7}$ | 0.78 (0.71–0.86) | rs179252 | **0.986** / **0.875** | 0.986–0.987 |
| FinnGen GO strict | 6,579 | rs2110695 ($1.34\times10^{-5}$) | $9.02\times10^{-5}$ | 0.82 (0.74–0.90) | rs179252 | **0.891** / 0.451 | 0.891–0.905 |

Disease lead, the strongest disease-associated variant among the matched cis variants, with its *P* value; rs179252 *P*, z-test *P* of the aligned G-allele disease association (the UKB source BOLT-LMM *P* was $1.3\times10^{-27}$). H4 top variant, the variant with the highest posterior under the shared-variant hypothesis; it is not necessarily the disease lead. PP.H4 under p12 = $10^{-5}$ and $10^{-6}$ (p1 = p2 = $10^{-4}$); bold, ≥0.80. Window range covers the full source cis-region, ±500 kb, ±250 kb and the five-outcome intersection at p12 = $10^{-5}$; it is not a confidence interval. All 240 window settings are in TSHR_windows. BBJ pairs European eQTL with East Asian disease statistics. CI, confidence interval; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Table S4. *TSHR* and *IGF1R* tissue expression extension

### A. Availability

| Gene / tissue | GTEx N | Instrument (F) | Status |
|---|---|---|---|
| *TSHR* / Thyroid | 574 | — | No instrument (minimum eQTL $P = 9.60\times10^{-8}$) |
| *IGF1R* / Thyroid | 574 | rs1815009 (F = 61.1) | Estimable |
| *TSHR* / Cultured fibroblasts | 483 | — | Not tested by GTEx in this tissue |
| *IGF1R* / Cultured fibroblasts | 483 | rs6598538 (F = 33.8) | Estimable |
| *TSHR* / Subcutaneous adipose | 581 | rs12101255 (F = 38.5) | Estimable |
| *IGF1R* / Subcutaneous adipose | 581 | — | No instrument (minimum eQTL $P = 3.56\times10^{-7}$) |
| *TSHR* / Visceral adipose | 469 | rs74064796 (F = 32.1) | Estimable |
| *IGF1R* / Visceral adipose | 469 | — | No instrument (minimum eQTL $P = 7.76\times10^{-6}$) |
| *TSHR* / Whole blood | 670 | — | No instrument (minimum eQTL $P = 2.46\times10^{-4}$) |
| *IGF1R* / Whole blood | 670 | — | No instrument (minimum eQTL $P = 5.42\times10^{-5}$) |

## B. Estimates

| Gene / tissue | Dataset | Family | OR (95% CI) | *P* value | Below family threshold | Direction vs blood |
|---|---|---|---|---|---|---|
| *TSHR* / Subcutaneous adipose | BBJ GD | Secondary | 0.46 (0.38–0.55) | $1.92\times10^{-15}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | FinnGen GD | Primary | 0.36 (0.31–0.43) | $1.28\times10^{-34}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | UKB hyperthyroidism | Secondary | 0.31 (0.27–0.37) | $2.86\times10^{-42}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | FinnGen GO | Secondary | 0.45 (0.32–0.63) | $5.76\times10^{-6}$ | Yes | Same |
| *TSHR* / Subcutaneous adipose | FinnGen GO strict | Secondary | 0.51 (0.35–0.74) | $3.67\times10^{-4}$ | Yes | Same |
| *TSHR* / Visceral adipose | BBJ GD | Secondary | 0.56 (0.46–0.67) | $2.64\times10^{-10}$ | Yes | Same |
| *TSHR* / Visceral adipose | FinnGen GD | Primary | 0.50 (0.43–0.58) | $4.09\times10^{-19}$ | Yes | Same |
| *TSHR* / Visceral adipose | UKB hyperthyroidism | Secondary | 0.39 (0.34–0.45) | $9.19\times10^{-38}$ | Yes | Same |
| *TSHR* / Visceral adipose | FinnGen GO | Secondary | 0.69 (0.49–0.98) | 0.0368 | No | Same |
| *TSHR* / Visceral adipose | FinnGen GO strict | Secondary | 0.78 (0.54–1.13) | 0.186 | No | Same |
| *IGF1R* / Thyroid | BBJ GD | Secondary | 1.23 (0.90–1.68) | 0.200 | No | Same |
| *IGF1R* / Thyroid | FinnGen GD | Primary | 0.76 (0.56–1.02) | 0.0692 | No | Opposite |
| *IGF1R* / Thyroid | UKB hyperthyroidism | Secondary | 1.33 (0.99–1.80) | 0.0594 | No | Same |
| *IGF1R* / Thyroid | FinnGen GO | Secondary | 1.02 (0.54–1.94) | 0.945 | No | Same |
| *IGF1R* / Thyroid | FinnGen GO strict | Secondary | 1.23 (0.62–2.44) | 0.549 | No | Same |
| *IGF1R* / Cultured fibroblasts | BBJ GD | Secondary | 0.91 (0.70–1.20) | 0.511 | No | Opposite |
| *IGF1R* / Cultured fibroblasts | FinnGen GD | Primary | 1.11 (0.89–1.38) | 0.359 | No | Same |
| *IGF1R* / Cultured fibroblasts | UKB hyperthyroidism | Secondary | 1.03 (0.82–1.29) | 0.815 | No | Same |
| *IGF1R* / Cultured fibroblasts | FinnGen GO | Secondary | 0.78 (0.49–1.25) | 0.301 | No | Opposite |
| *IGF1R* / Cultured fibroblasts | FinnGen GO strict | Secondary | 0.78 (0.47–1.30) | 0.341 | No | Opposite |

Post hoc fixed-panel extension using GTEx v8; tissue colocalization was not performed. A, availability of the ten gene–tissue combinations; F statistics are for the single retained instrument. ‘Not tested’ means the gene is not among the genes GTEx tested in that tissue. GTEx whole blood (n = 670) yielded no instrument at the fixed threshold, whereas the main analysis used eQTLGen (up to 31,684 participants). B, all 20 estimates. The primary family is the ten FinnGen GD comparisons ($P < 0.005$); the secondary family is the 40 comparisons with the other datasets ($P < 0.00125$); unestimable combinations stay in both denominators. ORs are per unit of inverse-normal-transformed tissue expression and are not comparable in size with the blood ORs. With one instrument, each *P* value equals that variant’s disease-association *P* value. The adipose *TSHR* instruments (rs12101255, rs74064796; 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and re-express the regional association rather than providing independent evidence. Cultured fibroblasts and adipose tissue are not orbital tissue. Direction, compared with the blood estimate in the same dataset. CI, confidence interval; GTEx, Genotype-Tissue Expression; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

## Figure S1

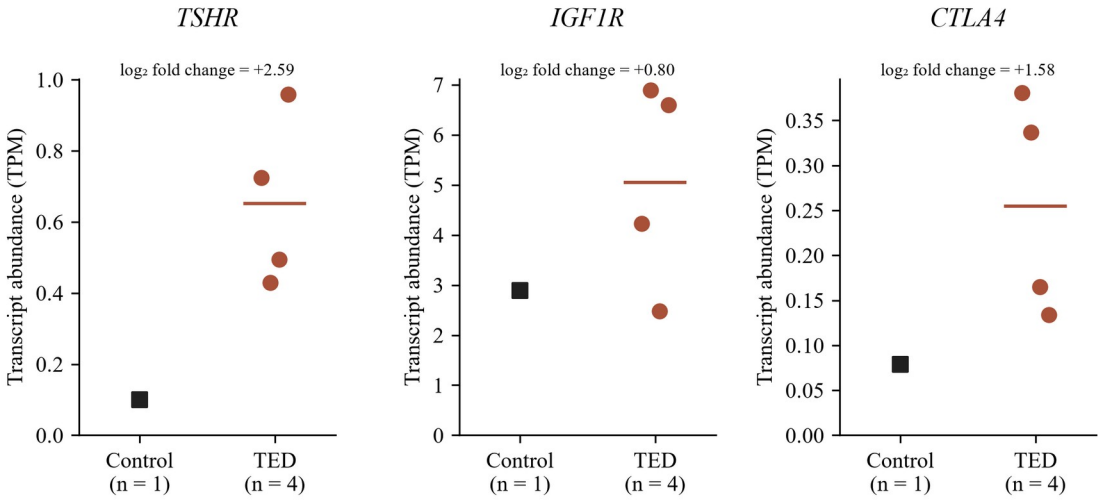



**Figure S1. Descriptive orbital expression of *TSHR*, *IGF1R* and *CTLA4*.** Transcript abundance in orbital adipose tissue collected during orbital decompression from four patients with inactive TED, and in nasal fat pad from one person without thyroid disease undergoing cosmetic upper blepharoplasty. Points are biological samples (technical replicates combined); horizontal lines show TED means. Fold change is $\log_2$[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Vertical axes differ by gene. The control came from a different fat depot, and one control cannot estimate control-group variance; no inferential test was performed, and these observations do not validate the genetic findings. TED, thyroid eye disease; TPM, transcripts per million.

## Figure S2

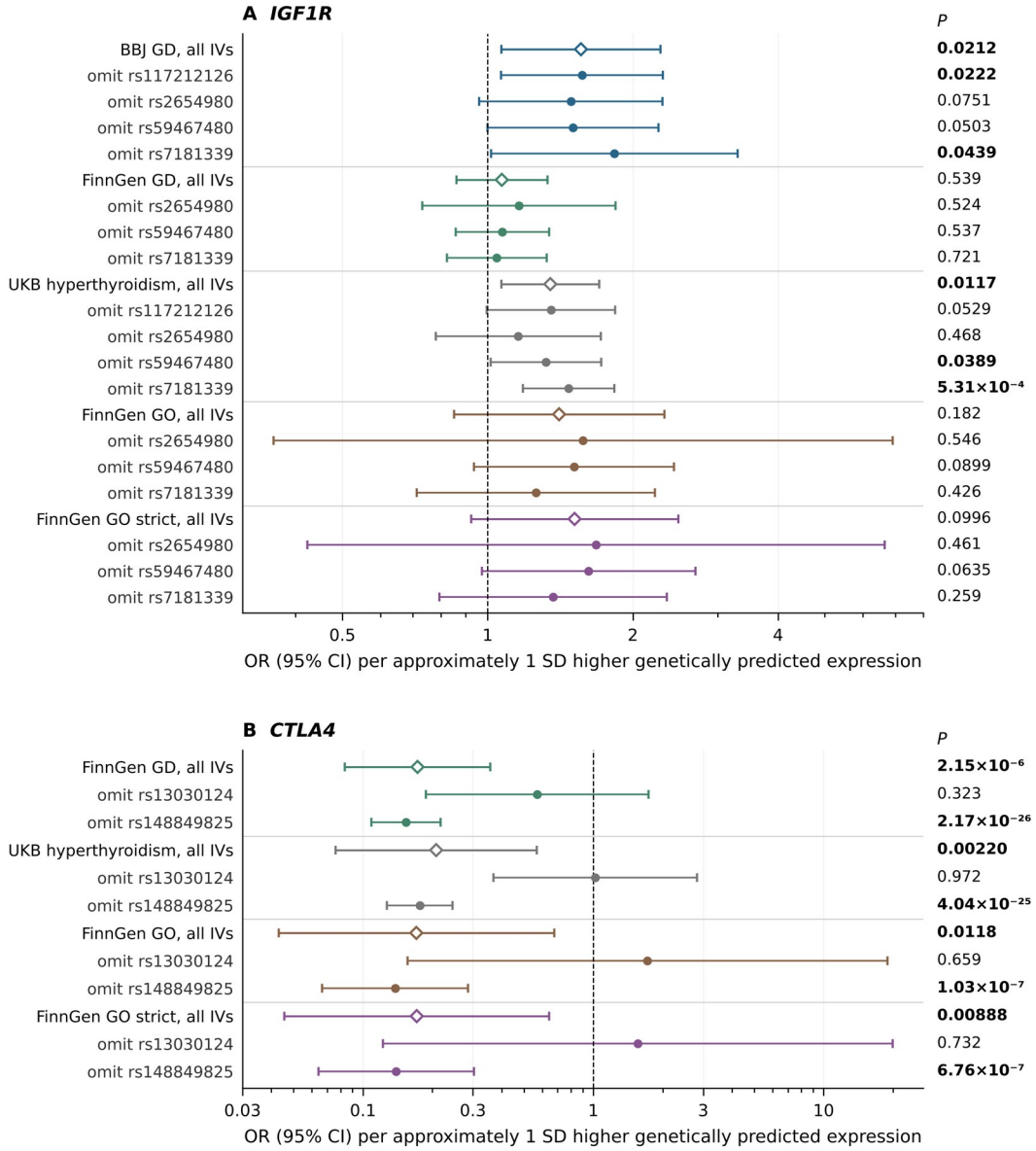



**Figure S2. Dependence of *IGF1R* and *CTLA4* estimates on individual instruments.** For each dataset, the open diamond is the estimate using all instruments (as in Table 2), and the filled points below it omit the named instrument (IVW when at least two instruments remain, Wald ratio when one remains). Lines are 95% CIs on a logarithmic axis; the dashed line is OR = 1. *P* values are shown at right; bold, nominal $P < 0.05$. (A) *IGF1R* in all five datasets. (B) *CTLA4* in the four European datasets; BBJ *CTLA4* and all *TSHR* estimates use one instrument and could not be assessed. The 25 post hoc omission estimates reuse the same association data and are not independent replications; loss of nominal significance can reflect reduced precision as well as the omitted variant's contribution. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.